(ii) Regression Equation of *Y* on *X*;

$Y-\overline{Y} = b_{yx}(X-\overline{X})$

where $\overline{X}$ is the mean of *X* series,

$\overline{Y}$ is the mean of *Y* series,

$b_{yx} = r\frac{\sigma y}{\sigma x} = \frac{\Sigma xy}{\Sigma x^2}$ is known as the regression coefficient of *Y* on *X*, $x = (X-\overline{X})$ and $y = (Y-\overline{Y})$

Note: Instead of finding out the values of correlation coefficient σ_x, σ_y, etc, we can find the value of regression coefficient by calculating Σxy and Σy^2.

Case 3: Deviations taken from Assumed Mean

When actual means of *X* and *Y* variables are in fractions the calculations can be simplified by taking the deviations from the assumed means. The regression equations and their coefficients are written as follows

(i) Regression Equation of Y on X:

$$Y-\overline{Y} = b_{yx}(X-\overline{X})$$

$$b_{yx} = \frac{N\Sigma dxdy-(\Sigma dx)(\Sigma dy)}{N\Sigma dx^2-(\Sigma dx)^2}$$

(ii) Regression Equation of X on Y:

$$X-\overline{X} = b_{xy}(Y-\overline{Y})$$

$$b_{xy} = \frac{N\Sigma dxdy-(\Sigma dx)(\Sigma dy)}{N\Sigma dy^2-(\Sigma dy)^2}$$

where $dx = X-A$, $dy = Y-B$, *A* and *B* are assumed means or arbitrary values are taken from *X* and *Y* respectively.

Properties of Regression Coefficients

(i) Correlation Coefficient is the geometric mean between the regression coefficients $r = \pm\sqrt{b_{xy}\times b_{yx}}$.

(ii) If one of the regression coefficients is greater than unity, the other must be less than unity.

(iii) Both the regression coefficients are of same sign.

Example 9.9

Calculate the regression coefficient and obtain the lines of regression for the following data

X	1	2	3	4	5	6	7
Y	9	8	10	12	11	13	14

Solution:

X	Y	X^2	Y^2	XY
1	9	1	81	9
2	8	4	64	16
3	10	9	100	30
4	12	16	144	48
5	11	25	121	55
6	13	36	169	78
7	14	49	196	98
$\Sigma X = 20$	$\Sigma Y = 77$	$\Sigma X^2 = 140$	$\Sigma Y^2 = 875$	$\Sigma XY = 334$

Table 9.7

$\overline{X} = \frac{\Sigma X}{N} = \frac{28}{7} = 4$, $\overline{Y} = \frac{\Sigma Y}{N} = \frac{77}{7} = 11$

Regression coefficient of *X* on *Y*:

$$b_{xy} = \frac{N\Sigma XY-(\Sigma X)(\Sigma Y)}{N\Sigma Y^2-(\Sigma Y)^2}$$

$$= \frac{7(334)-(28)(77)}{7(875)-(77)^2}$$

$$= \frac{2338-2156}{6125-5929} = \frac{182}{196}$$

$\therefore b_{xy} = 0.929$

(i) Regression equation of *X* on *Y*:

$X-\overline{X} = b_{xy}(Y-\overline{Y})$

$X-4 = 0.929(Y-11)$

$X-4 = 0.929Y-10.219$

$\therefore$ The regression equation *X* on *Y* is $X = 0.929Y-6.219$.

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